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Section: BSIT – 4C

Installation Guide

I. Project Overview

Operating Systems serve as the foundation of every enterprise IT infrastructure. Before a server can host websites, databases, cloud services, or enterprise applications, it must first be properly installed, configured, secured, and documented.

In this project, you will assume the role of a **Junior System Administrator** responsible for deploying a new Ubuntu Server for a startup company. You will perform a complete operating system installation, configure essential settings, verify the installation, compare modern boot technologies, and create professional documentation suitable for future administrators.

The deliverables from this project will become part of your professional GitHub portfolio.

II. Learning Objectives

After completing this project, students should be able to:

Knowledge

Students should be able to:

1. Explain the purpose of an operating system in enterprise environments.
2. Differentiate BIOS and UEFI firmware.
3. Explain the stages of the computer boot process.
4. Compare Ubuntu Server, Windows Server, and Rocky Linux.

Skills

Students should be able to:

- Install Ubuntu Server in a virtual machine.
- Configure server settings during installation.
- Enable secure remote administration using SSH.
- Verify server functionality.
- Document installation procedures.
- Produce professional technical documentation.

III. Software Requirements

The following software and installation media were required to create, install, configure, and verify the server operating systems used in this project.

- Oracle VirtualBox or VMware Workstation
- Ubuntu Server LTS ISO
- Windows Server Evaluation ISO

IV. Virtual Machine Specifications

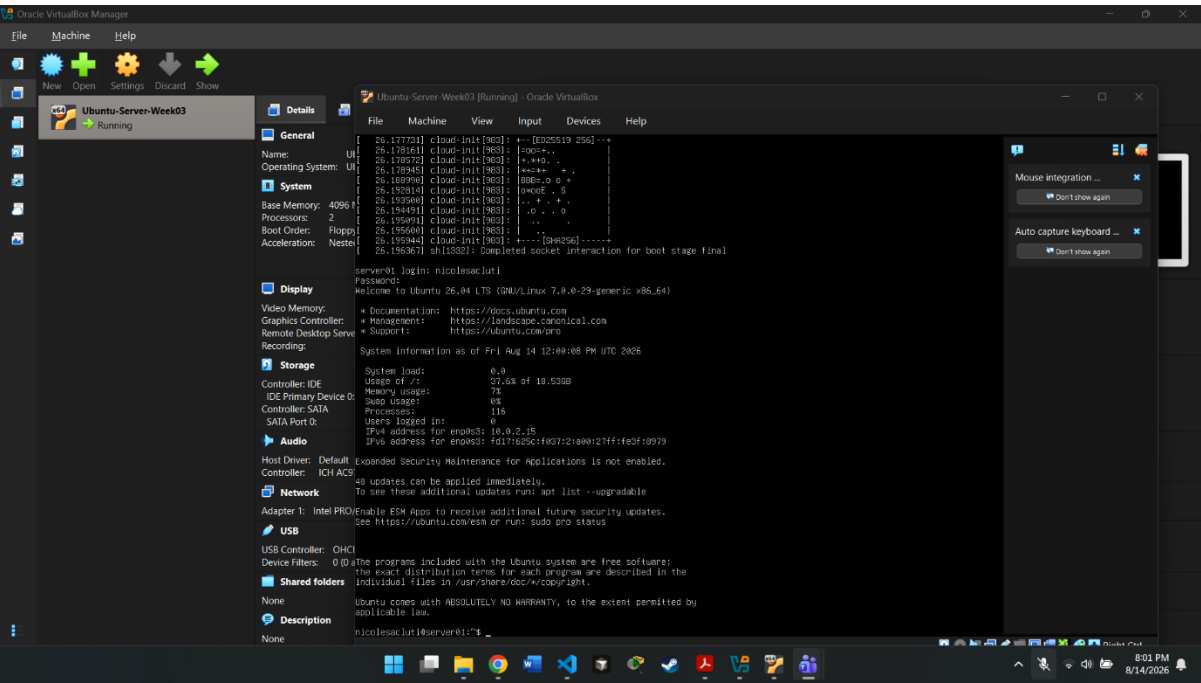
The virtual machines were configured with the following hardware resources to provide a suitable environment for installing and running the server operating systems.

Component	Minimum Requirement
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	Virtual Processors
Storage	40 GB (VDI/VMDK)
Network	NAT
Optical Drive	Ubuntu Server ISO

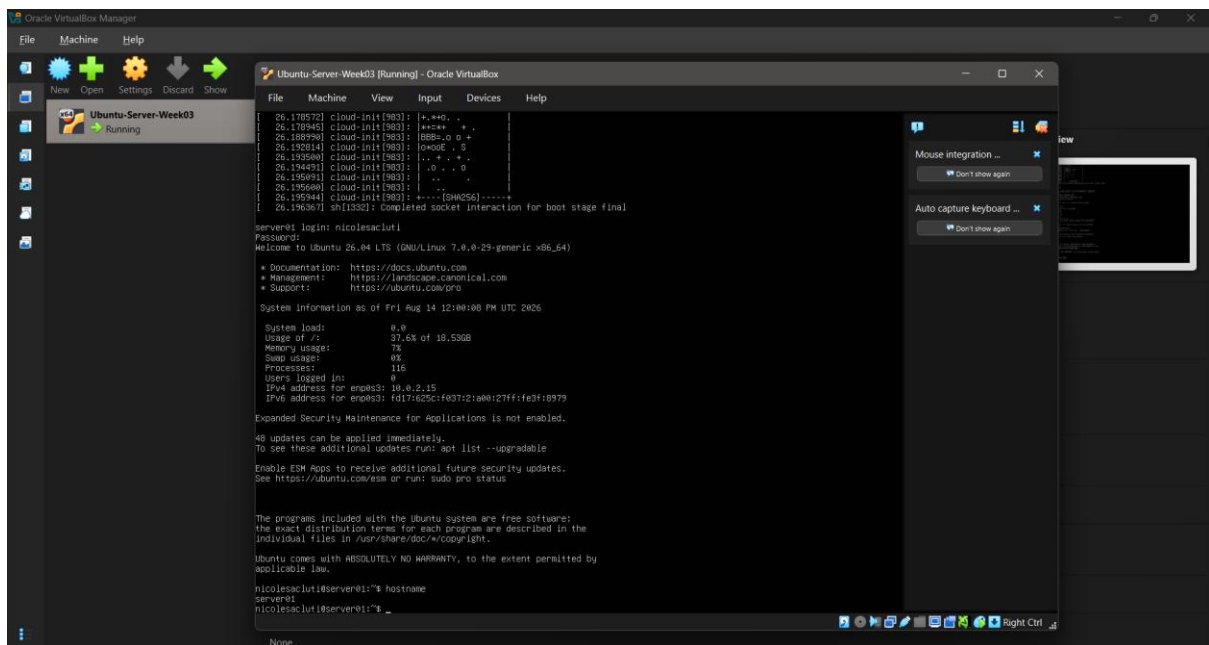
V. Step-by-Step Installation Procedure

The following procedure presents the step-by-step process followed to install and configure the Ubuntu Server virtual machine using Oracle VirtualBox.

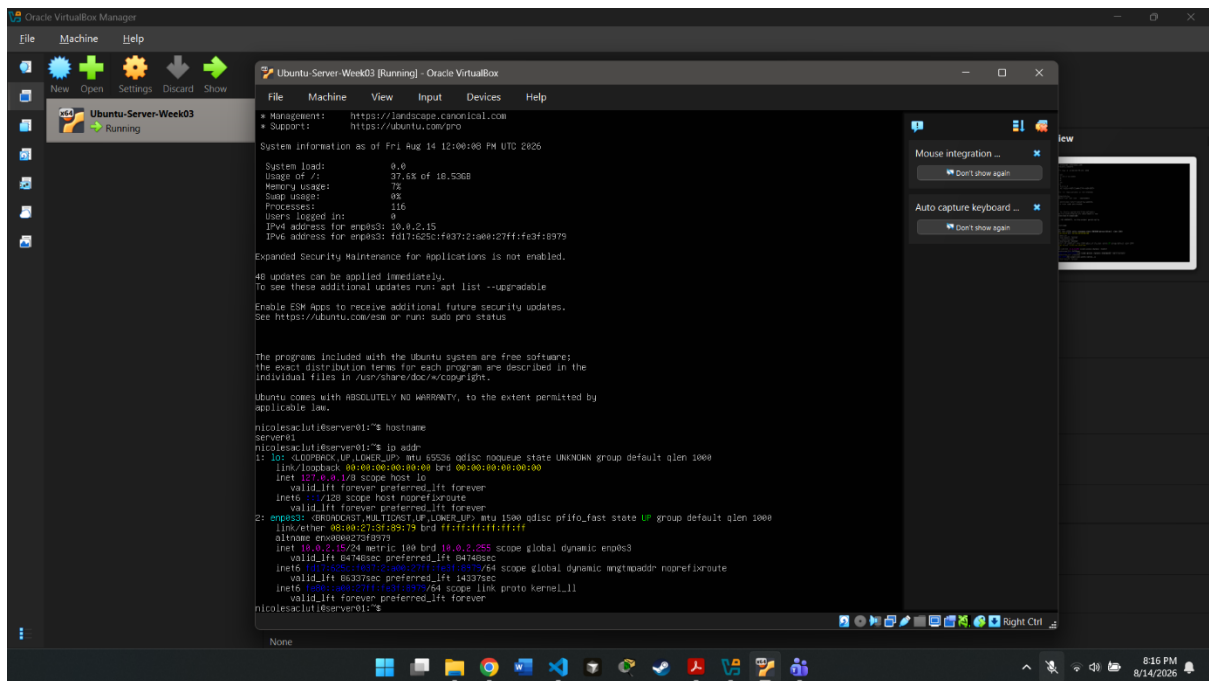
Task 1 – Login



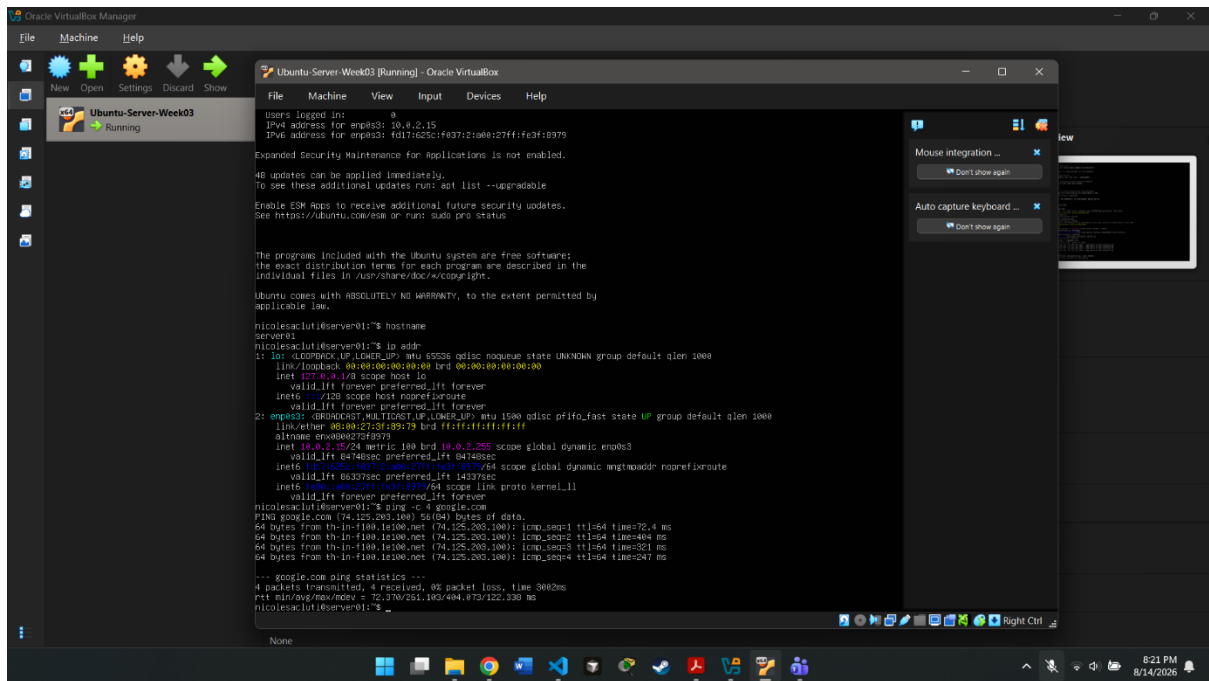
Task 2 – Verify Hostname



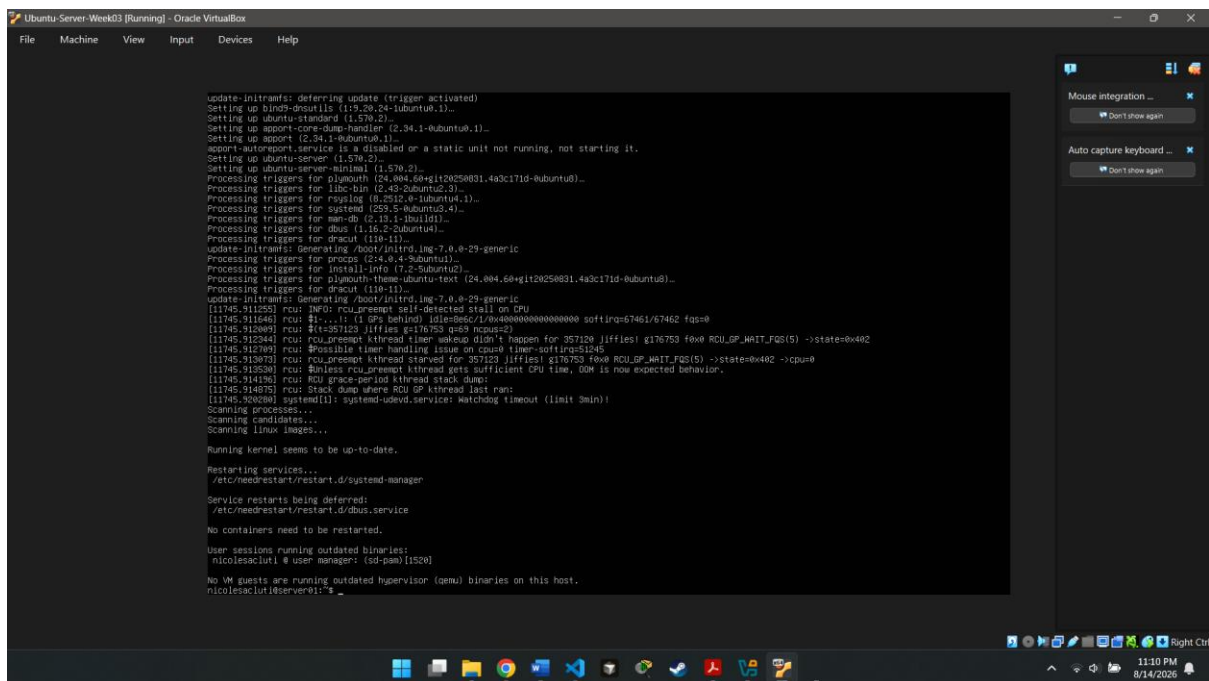
Task 3 – Verify IP Address



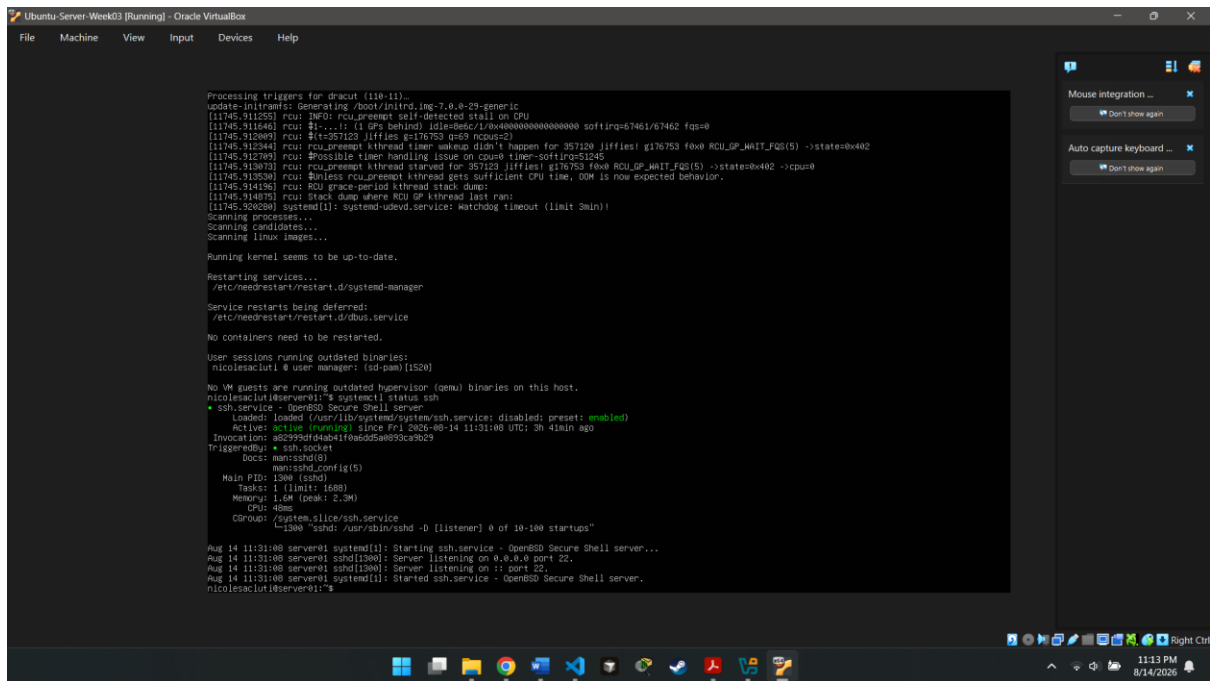
Task 4 – Test Internet Connectivity



Task 5 – Update the Server



Task 6 – Verify SSH Service



VI. BIOS vs UEFI Comparison

BIOS and UEFI are firmware technologies responsible for initializing computer hardware and starting the operating system, with UEFI providing several improvements over traditional BIOS.

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is older firmware that initializes the computer's hardware and starts the operating system.	UEFI (Unified Extensible Firmware Interface) is modern firmware that replaces the traditional BIOS and provides more advanced boot and hardware management features.
Boot Process	BIOS performs hardware checks, looks for a bootable device, and loads the bootloader from the disk's boot sector.	UEFI initializes hardware and loads a bootloader from the EFI System Partition. It can also verify boot software using Secure Boot.
Maximum Disk Support	Traditional BIOS commonly works with MBR, which has a practical disk-size limit of about 2 TB for standard 512-byte sectors.	UEFI commonly uses GPT, which supports disks much larger than 2 TB. GPT can support extremely large disks and many partitions.
Partition Style	Mainly uses MBR (Master Boot Record) .	Mainly uses GPT (GUID Partition Table) .
Security Features	Has limited built-in boot security compared with UEFI. Traditional	Supports security features such as Secure Boot , which checks that

	BIOS does not provide Secure Boot.	trusted software is used during the boot process.
Speed	Generally has a slower and more basic boot process.	Generally provides faster boot and resume times.
Advantages	Simple, widely compatible with older hardware and operating systems, and useful for legacy systems.	Supports large drives, more partitions, faster booting, modern security features, and a more flexible firmware environment.
Disadvantages	Limited partitioning and disk support, older technology, and fewer built-in security features.	May have compatibility issues with very old hardware or operating systems. It can also be more complex than traditional BIOS.
Modern Usage	Mostly found on older computers or used through legacy/compatibility modes.	Used on most modern computers and is the standard firmware environment for current systems.

BIOS and UEFI both perform an important job when a computer starts. They initialize the computer's hardware and then begin the process of loading the operating system. However, BIOS is an older technology that was designed for computers with much simpler hardware and storage requirements. UEFI was developed as a replacement for the traditional BIOS and provides features that are better suited to modern computers. Microsoft describes UEFI as the replacement for the older BIOS firmware interface.

One major difference between BIOS and UEFI is how they handle disk partitions. Traditional BIOS normally works with the MBR partition style. MBR has limitations on disk size and the number of primary partitions. UEFI normally works with GPT, which supports much larger drives and many more partitions. The UEFI specification also explains that GPT provides advantages such as support for many partitions, backup partition tables, and improved data integrity through CRC32 checks.

UEFI also provides better security. One of its most important features is Secure Boot. Secure Boot checks the digital signatures of boot software and allows the computer to start only trusted software. This helps protect the computer from certain types of bootkits and other malware that try to load before the operating system.

Another advantage of UEFI is improved boot performance. Modern UEFI systems can provide faster boot and resume times than older BIOS-based systems. UEFI also provides a more flexible environment for manufacturers to add features and tools. Modern UEFI interfaces can include graphical menus and support mouse navigation, although the exact features depend on the computer manufacturer.

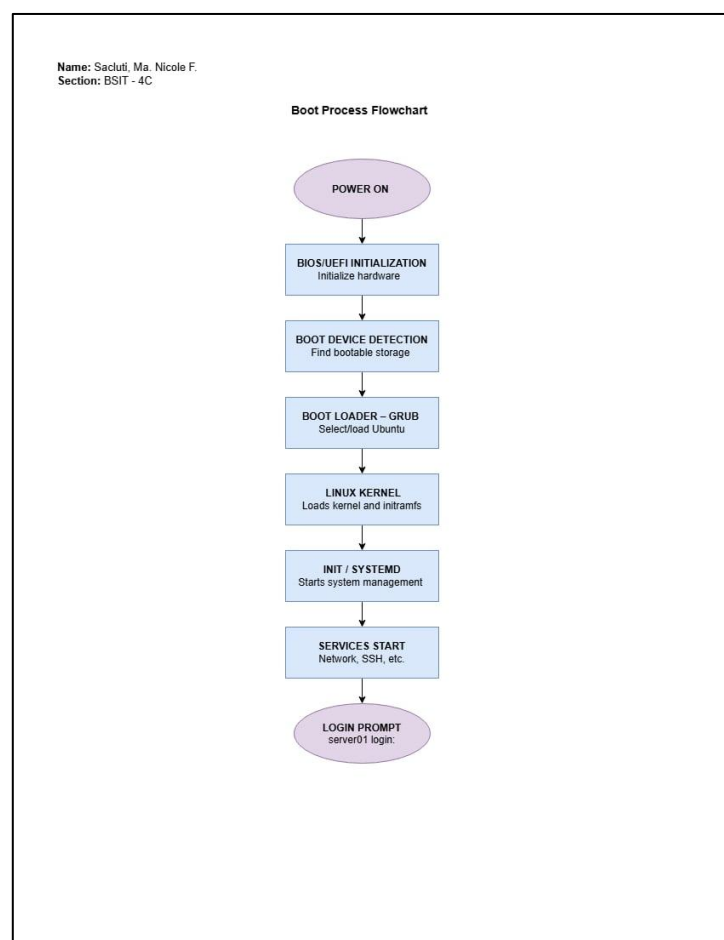
BIOS is still useful for older computers and legacy operating systems because of its compatibility. Some modern computers also provide a compatibility mode that allows older BIOS-style booting. However, new operating systems and modern computer hardware are

designed mainly around UEFI and GPT. For example, Microsoft requires UEFI and GPT for Windows installations using the modern UEFI boot mode.

Overall, UEFI has largely replaced BIOS because it is better suited to modern computing. It supports larger storage devices, more partitions, faster booting, and stronger security features such as Secure Boot. BIOS remains important for understanding older computer systems, but UEFI is the preferred firmware technology for modern computers because it provides greater flexibility, compatibility with newer hardware, and improved security.

VII. Boot Process Flowchart

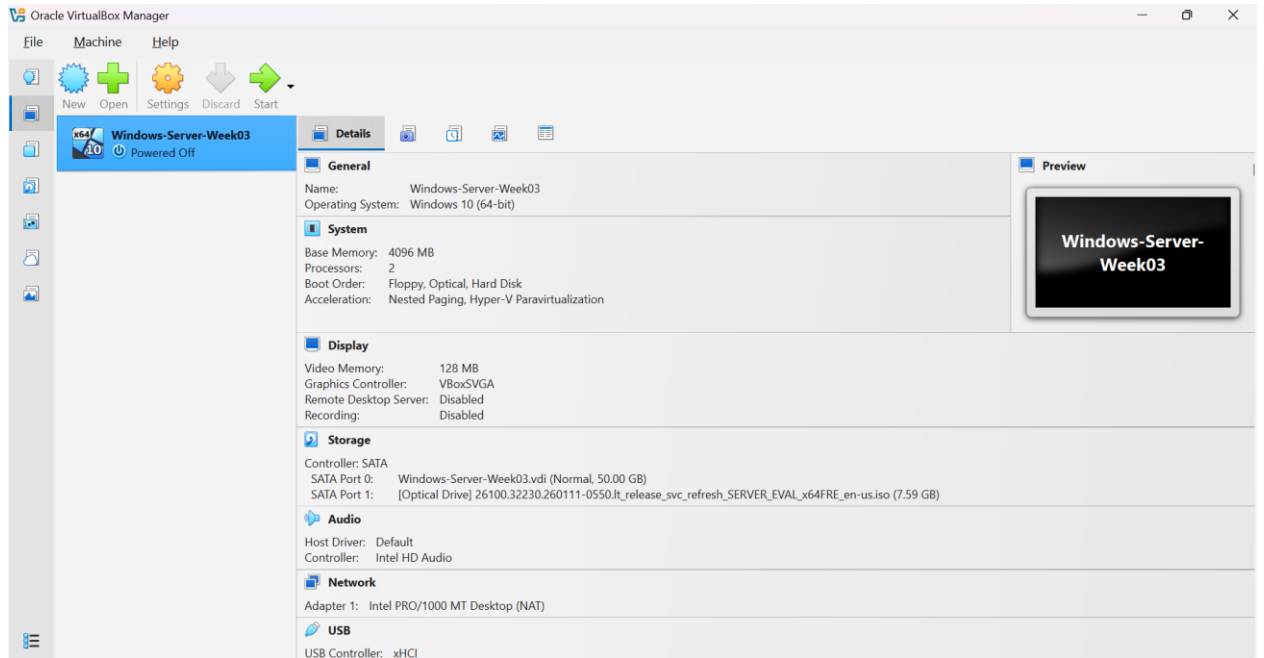
The flowchart illustrates the major stages of the Ubuntu Server boot process, from powering on the computer to reaching the login prompt.



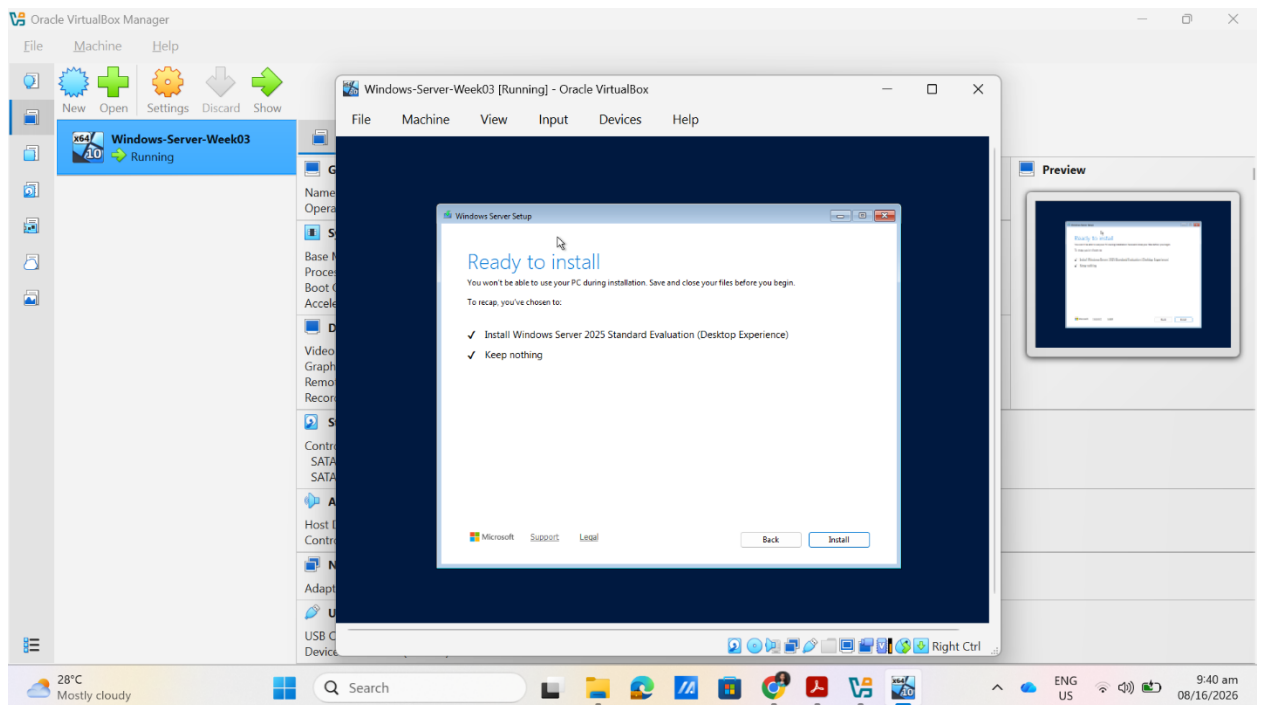
VIII. Windows Server Installation Summary

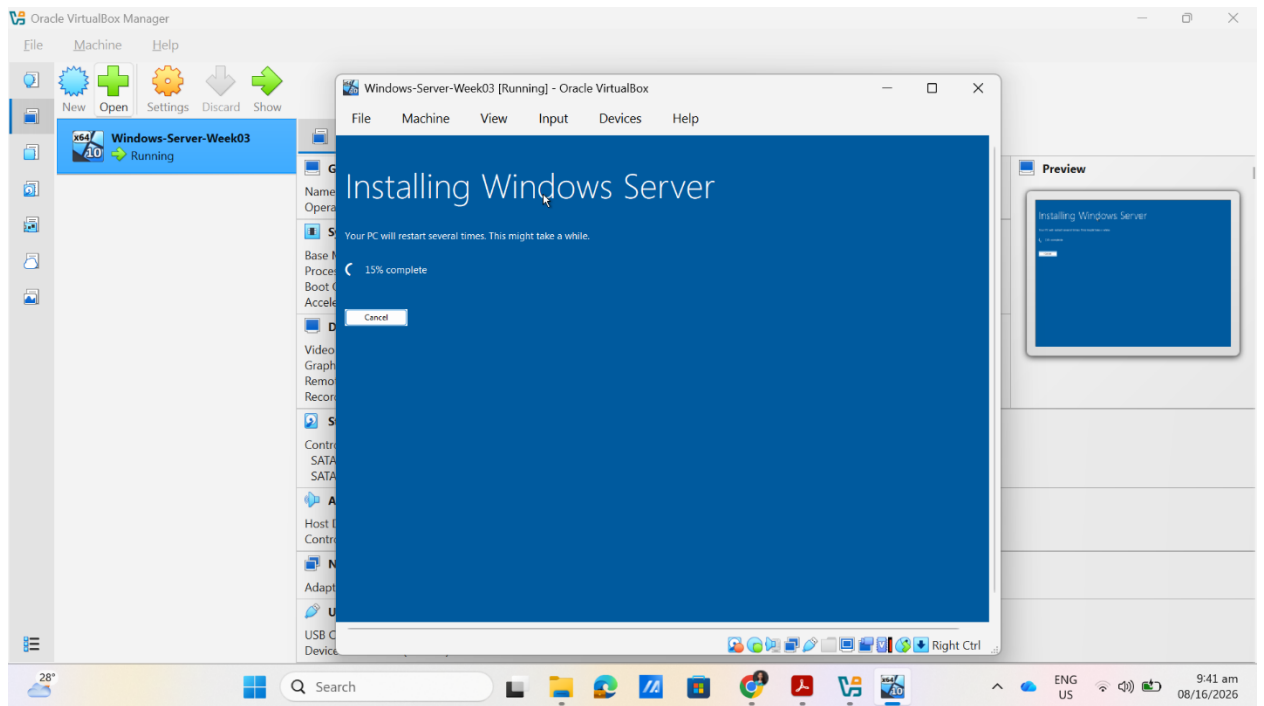
The Windows Server Evaluation installation was performed in a separate virtual machine to provide practical experience with the deployment and initial configuration of a Windows-based server operating system.

Task 1 – Create the VM

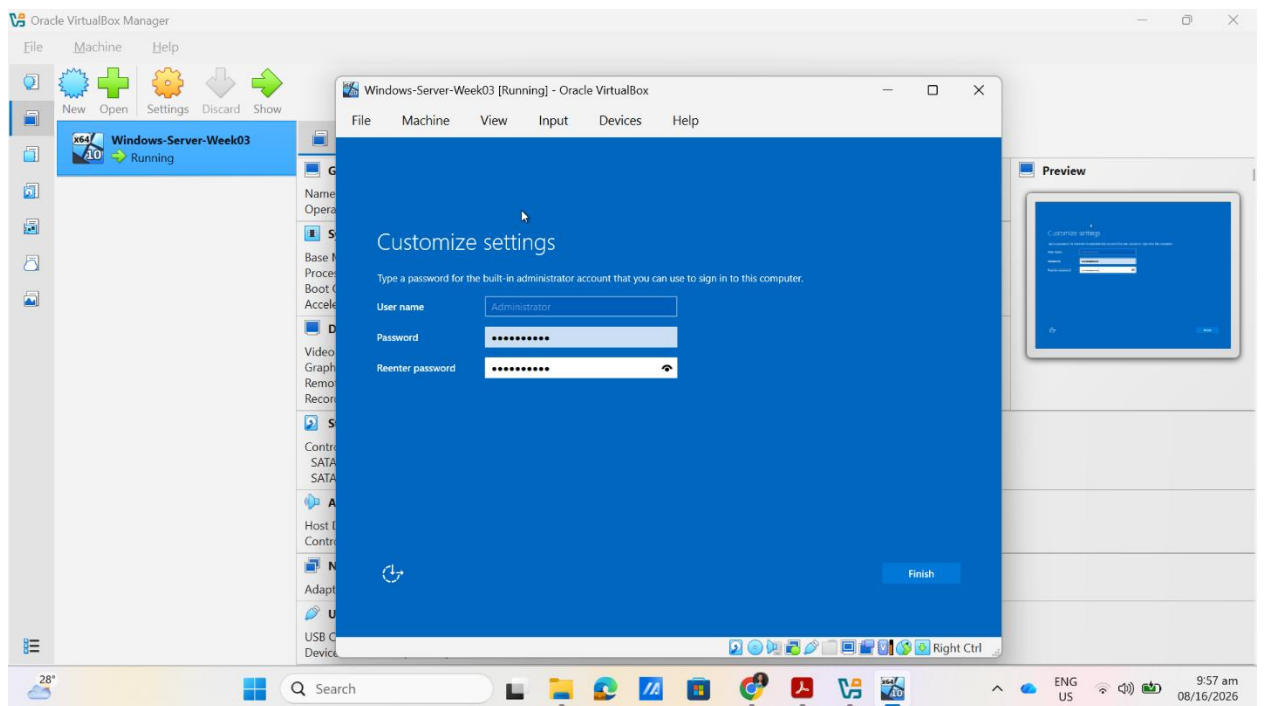


Task 2 – Windows Server Installation

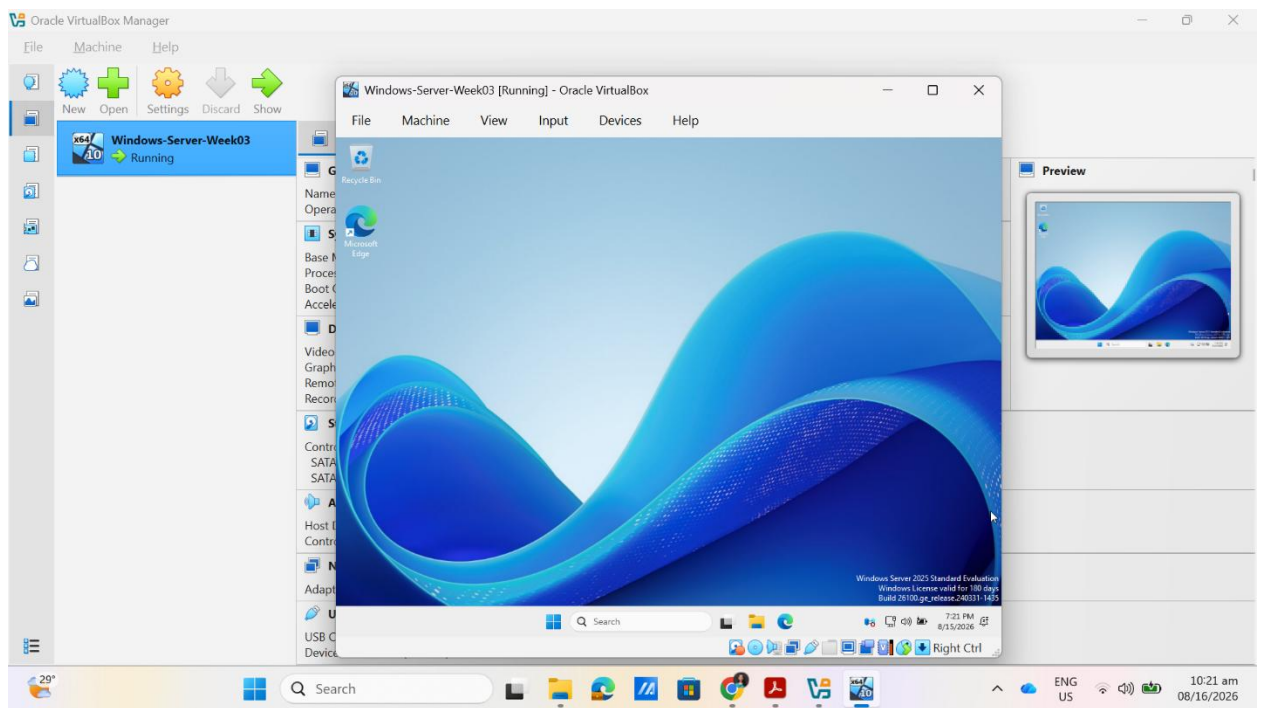
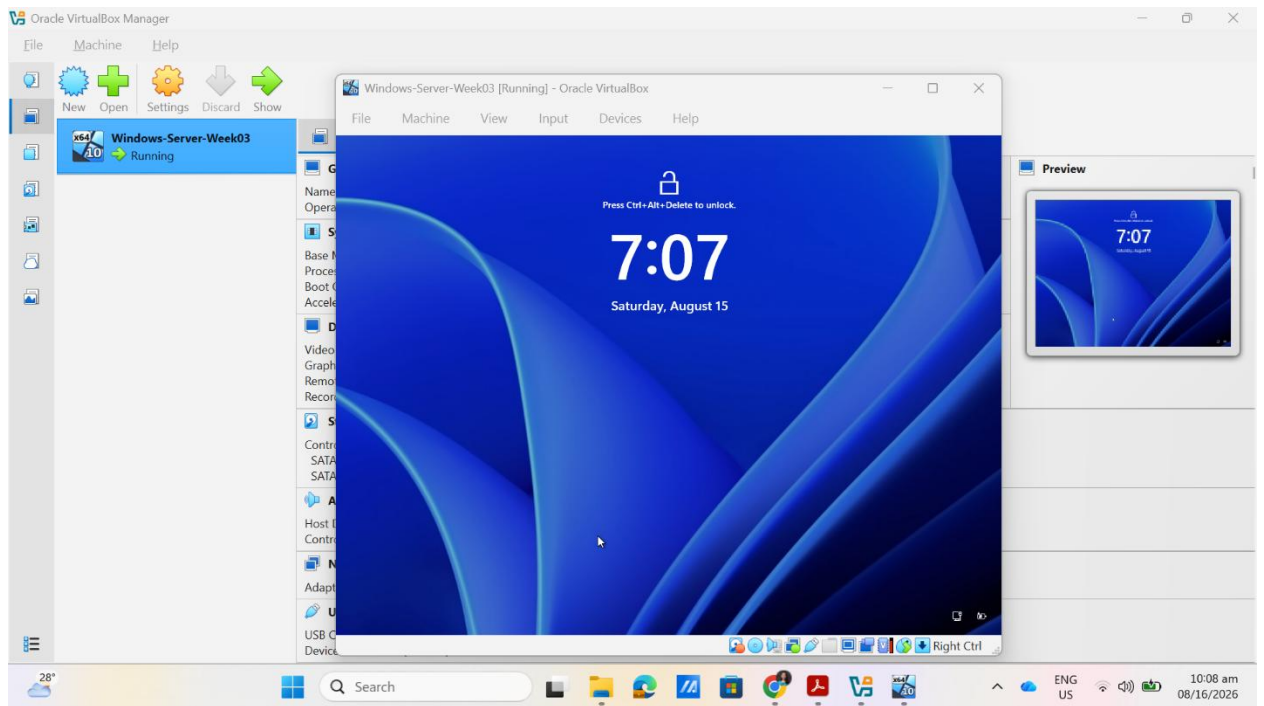




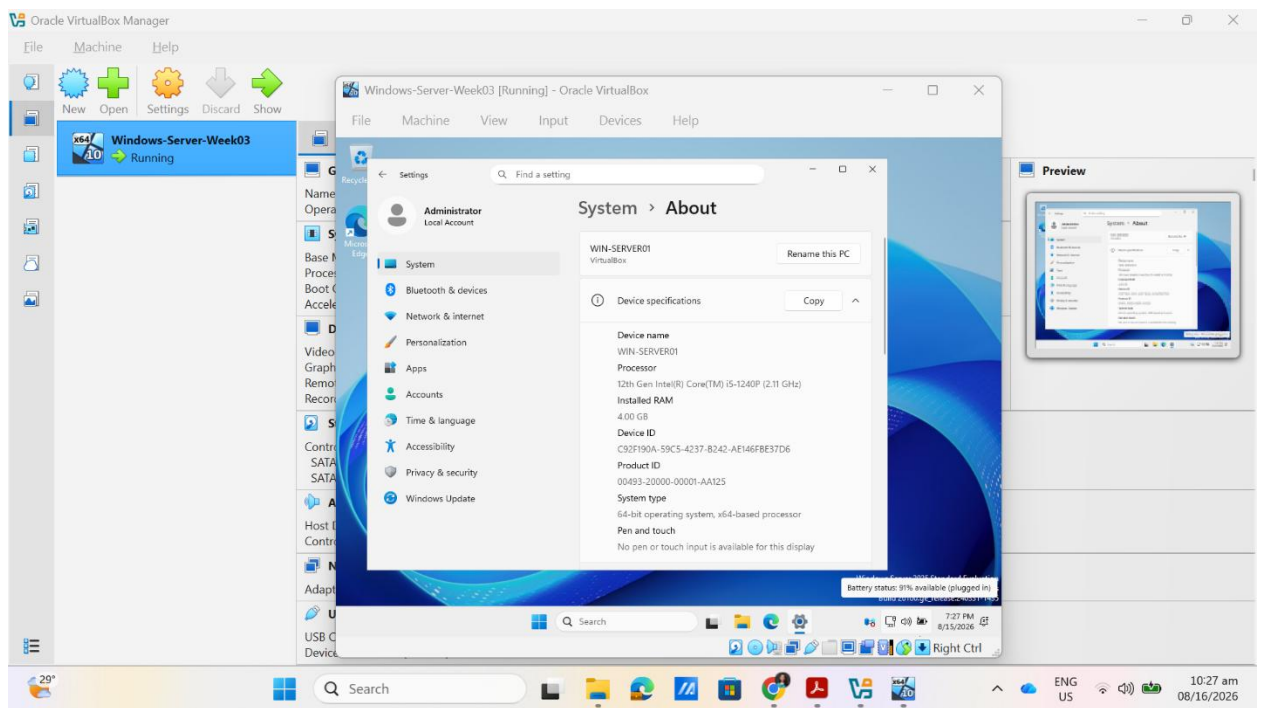
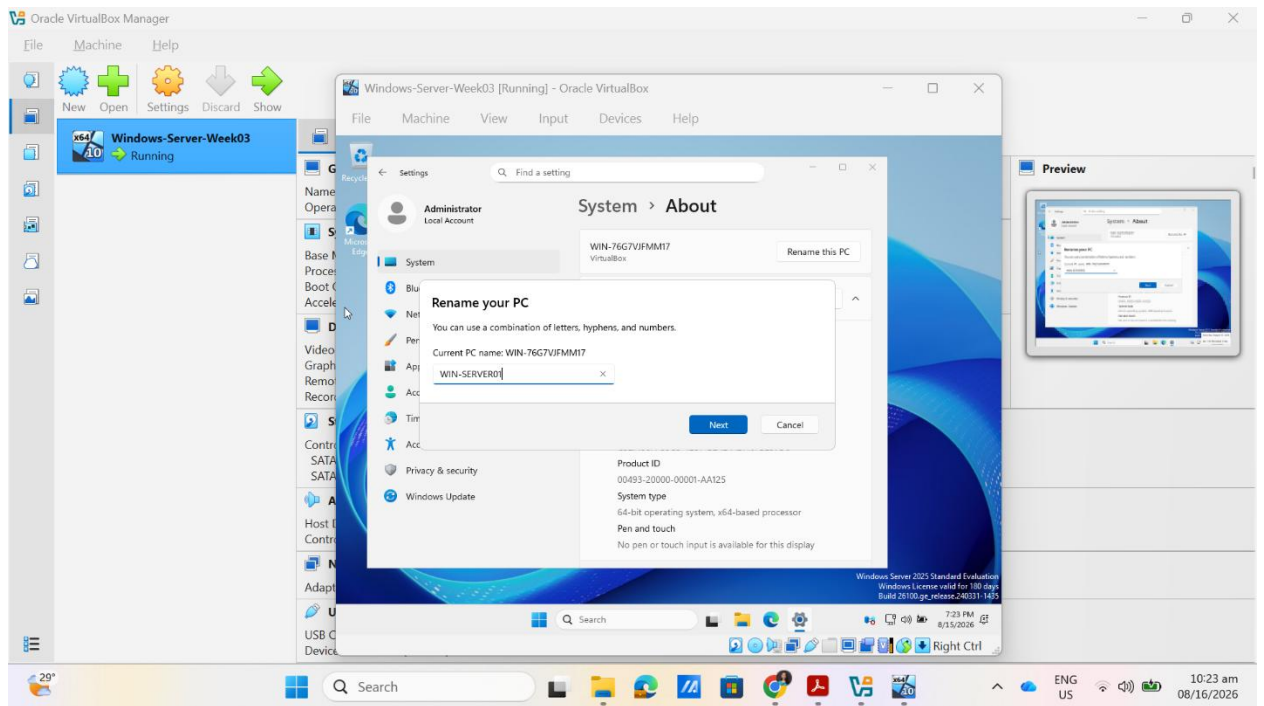
Task 3 – Administrator Password



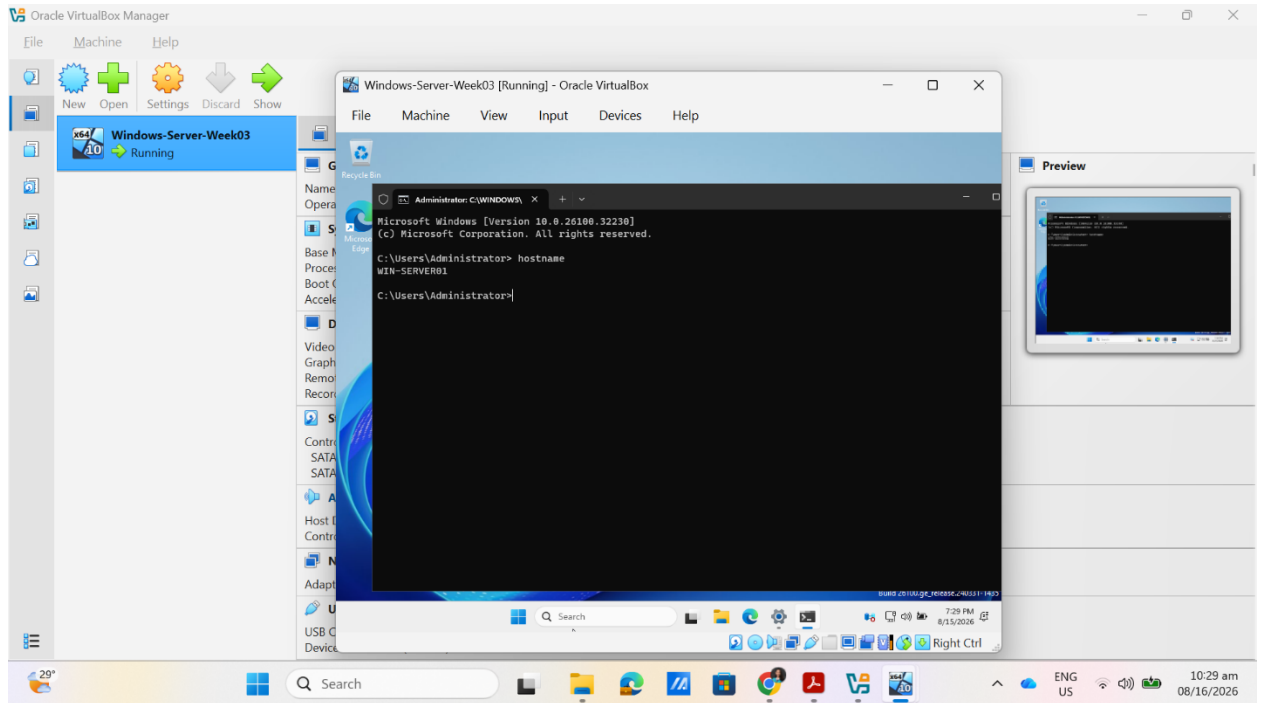
Task 4 – Successful Login



Task 5 – Assign Computer Name



Task 6 – Windows Server Verification



IX. OS Comparison Table

The following table compares Ubuntu Server, Windows Server, and Rocky Linux based on their features, administration methods, advantages, and common enterprise uses.

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary and requires a license for production use; evaluation editions are available for testing.	Open-source and freely available under various open-source licenses.	Open-source and freely available; designed as a downstream-compatible alternative to Red Hat Enterprise Linux (RHEL).
User Interface	Provides a graphical interface through Desktop Experience and also supports command-line administration through PowerShell and Command Prompt.	Primarily command-line based; graphical interfaces can be installed when needed.	Primarily command-line based; graphical environments can be installed if required.
Package Management	Uses Windows Update, Microsoft Store/installation packages, and tools such as PowerShell for software management.	Uses APT and .deb packages.	Uses DNF and RPM packages.

Security	Includes Windows Defender, Windows Firewall, security policies, Active Directory security controls, and other enterprise security features.	Uses Linux permissions, AppArmor, firewall tools such as UFW, SSH security, and regular security updates.	Uses SELinux, firewalld, Linux permissions, SSH security, and enterprise-oriented security tools.
Performance	Generally requires more system resources, particularly when using a graphical interface, but performs well for enterprise workloads.	Generally lightweight and efficient, making it suitable for servers with limited resources.	Efficient and stable, particularly for enterprise workloads requiring RHEL-compatible environments.
Typical Enterprise Use Cases	Active Directory, DNS, DHCP, file and print services, Microsoft applications, virtualization, and enterprise Windows environments.	Web servers, cloud infrastructure, containers, databases, development environments, networking, and general Linux server workloads.	Enterprise servers, web hosting, databases, virtualization, cloud infrastructure, and applications designed for RHEL-compatible environments.
Advantages	Strong Microsoft ecosystem integration, GUI administration, Active Directory, extensive enterprise management tools, and broad commercial software support.	Free and open-source, lightweight, flexible, large community, strong cloud/container support, and extensive software availability.	Free and open-source, stable, enterprise-focused, RHEL-compatible, long lifecycle, and strong security features.
Disadvantages	Licensing costs, higher resource requirements, and greater dependence on the Microsoft ecosystem.	May require more command-line knowledge, some commercial software may not be available natively, and configuration can be less familiar to Windows users.	Primarily command-line oriented, may have fewer beginner-friendly resources than Ubuntu, and some newer software may require additional configuration.

Windows Server, Ubuntu Server, and Rocky Linux are widely used server operating systems, but each is designed around different requirements and administrative environments. Windows Server is a proprietary operating system developed by Microsoft and is commonly used by organizations that rely heavily on Microsoft's ecosystem. It provides a graphical interface through the Desktop Experience option, while administrators can also use PowerShell and Command Prompt for command-line management. Windows Server is particularly suitable for environments that require services such as Active Directory, DNS, DHCP, file sharing, and

integration with other Microsoft products. Its major advantages are its user-friendly graphical administration tools, strong enterprise management capabilities, and integration with Microsoft technologies. However, licensing costs and higher resource requirements can make it more expensive to deploy than Linux-based alternatives.

Ubuntu Server is an open-source Linux distribution developed by Canonical and is available without traditional operating-system licensing costs. It is commonly administered through the command line and uses the APT package management system for installing and updating software. Ubuntu Server is widely used for web hosting, cloud computing, databases, containers, development environments, and network services. Its relatively lightweight design allows it to perform efficiently while using fewer resources than many graphical server environments. Ubuntu also has a large community and extensive documentation, making it a popular choice for both learning and professional server administration. However, administrators who are accustomed to graphical interfaces may initially find command-line administration more challenging.

Rocky Linux is another open-source Linux distribution designed to provide compatibility with Red Hat Enterprise Linux (RHEL). It uses DNF for package management and RPM as its package format. Rocky Linux is particularly appropriate for organizations that require a stable, enterprise-oriented Linux environment and compatibility with software designed for RHEL. Its security features include SELinux, firewall management, Linux file permissions, and secure remote administration through SSH. One of its major advantages is its enterprise focus, stability, and compatibility with the broader RHEL ecosystem. However, like Ubuntu Server, Rocky Linux is primarily command-line oriented, which can present a learning curve for beginners.

In terms of security, all three operating systems provide strong security capabilities but approach administration differently. Windows Server provides features such as Windows Defender, Windows Firewall, security policies, and Active Directory. Ubuntu Server provides Linux permissions, AppArmor, firewall configuration, SSH, and regular security updates, while Rocky Linux emphasizes enterprise security features such as SELinux and firewalld. Performance also depends heavily on the workload and configuration, but Linux servers are generally attractive for organizations seeking lightweight and efficient server environments.

Overall, there is no single operating system that is best for every enterprise. Windows Server is a strong choice for organizations built around Microsoft technologies, while Ubuntu Server is highly flexible for web, cloud, development, and general Linux workloads. Rocky Linux is particularly suitable for organizations seeking a stable, enterprise-oriented Linux platform with RHEL compatibility. The appropriate choice should therefore depend on the organization's existing infrastructure, technical requirements, available expertise, security requirements, budget, and intended server workloads.

X. Problems Encountered and Solutions Implemented

During the project, several challenges were encountered while installing and configuring the Ubuntu Server and Windows Server virtual machines. One of the initial problems was identifying the correct file to use when creating the virtual machine in Oracle VirtualBox. The Ubuntu Server ISO was located in the Downloads folder, while the virtual machine folder itself was initially empty, which caused confusion when selecting the installation media. This was resolved by selecting the downloaded Ubuntu Server ISO directly from the Downloads folder and attaching it as the virtual optical disk of the virtual machine.

Another problem occurred after the Ubuntu Server installation when the virtual optical disk could not be ejected normally. VirtualBox displayed a message stating that it was unable to eject the virtual optical disk and provided a Force Unmount option. The force unmount option was selected, and the virtual machine was allowed to continue until the installation process was completed and the server01 login: prompt appeared. This confirmed that the Ubuntu Server installation had successfully finished.

A problem was also encountered while updating Ubuntu Server. Running `sudo apt upgrade -y` initially produced a 403 Forbidden error when attempting to download packages from the regional Ubuntu mirror `ph.archive.ubuntu.com`. The problem was caused by the selected package mirror not allowing access to some of the required package files. The package source was changed from the Philippine mirror to the main Ubuntu archive mirror, `archive.ubuntu.com`. After changing the repository source, `sudo apt update` was executed successfully, and the system detected 53 packages that could be upgraded. The upgrade command was then executed again, allowing the available packages to be installed successfully.

During the Windows Server virtual machine setup, VirtualBox initially failed to boot the virtual machine because the Windows Server ISO was not properly attached as the bootable optical disk. The VirtualBox wizard also displayed a red warning when the ISO was selected during the unattended installation setup. To avoid the issue, the virtual machine was created and configured first without using the unattended installation option. The Windows Server Evaluation ISO was then attached manually through the virtual machine's optical drive settings. After restarting the virtual machine, Windows Server successfully loaded and displayed the Windows login screen.

These problems provided practical experience in virtual machine configuration, ISO management, package repository troubleshooting, operating system installation, and basic server administration. The troubleshooting process also demonstrated the importance of carefully reading error messages and verifying each configuration before proceeding to the next installation step.

XI. Lessons Learned

This project taught me that installing and managing a server operating system requires careful planning, attention to detail, and proper troubleshooting. I learned how to create and configure virtual machines using Oracle VirtualBox and how important it is to allocate

appropriate hardware resources such as memory, processors, and storage before beginning an installation. I also gained practical experience installing Ubuntu Server and Windows Server, configuring hostnames and administrator accounts, setting up network connectivity, and enabling SSH for remote administration.

I learned that errors during installation and system updates are normal and can often be resolved by carefully reading the error messages and identifying their causes. For example, troubleshooting the Ubuntu package repository issue helped me understand how software repositories affect system updates. I also learned the importance of verifying configurations after installation instead of assuming that everything was configured correctly.

Another important lesson was the value of technical documentation. Recording installation steps, commands, screenshots, problems, and solutions makes it easier for other administrators to understand and reproduce the same setup. Overall, the project improved my confidence in working with server operating systems and strengthened my problem-solving, troubleshooting, documentation, and system administration skills.

XII. References

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